

Answer any two from question no 1:

5*2=10

1.

- a. A computer-controlled robotic arm needs to reach a target angle of $\theta = 0.523$ rad (30°). The nonlinear equation is

$$f(\theta) = \theta \cos \theta - 0.2 = 0.$$

Start $\theta_0 = 0.5$. Do two iterations; correct to 3 d.p..

- b. You are training a machine learning model with cost function:

$$J(w) = (w - 3)^2$$

Start with $w_0 = 0$, learning rate $\eta = 0.1$.

- i. Perform three iterations of SGD update:

$$w_{new} = w_{old} - \eta \frac{dJ}{dw}$$

- ii. Plot or tabulate the sequence of w values.

- iii. Explain in your own words why SGD is useful for large datasets.

- c. A dataset records whether students pass (1) or fail (0) based on study hours:

Hours (X)	1	2	3	4	5
Pass/Fail (Y)	0	0	0	1	1

- i. Explain why linear regression is not suitable here.

- ii. Write the logistic regression model equation.

- iii. Using $P(Y = 1|X) = \frac{1}{1 + e^{-(a+bX)}}$, explain how parameters a and b affect the decision boundary.

2. Assignment related paper presentation.

5

Mid Term Examination of B. Sc. Engg. (CSE) January-June 2024
Course Title: Computer Organization and Architecture Course Code : CTF-313

Time: 50 Min

Marks: 15

N.B. Answer the following questions. (Split answers are highly discouraged)

- 1 Why need to study the computer organization and architecture course for CSE degree?
- 2 What is computer family? Give the evaluation of intel processors.
- 3 On the IAS, describe the process that the CPU must undertake to read a value from memory and to write a value to memory in terms of what is put into the MAR, MBR, address bus, data bus, and control bus.
- 4 Define memory cycle time and transfer rate. How does the multilevel cache work in computer system.
- 5 Consider a machine with a byte addressable main memory of 2^{16} bytes and block size of 8 bytes. Assume that a direct mapped cache consisting of 32 lines is used with this machine.
 - i. How is a 16-bit memory address divided into tag, line number and byte number?
 - ii. Into what line would bytes with each of the following addresses be stored?
0001 0001 0001 1011
 - iii. Suppose the byte with address 0001 1010 0001 1010 is stored in the cache. What are the addresses of the other bytes stored along with it?
 - iv. How many total bytes of memory can be stored in the cache?

5th Semester (Level-3, Semester-I), **Mid-Term Exam** of B.Sc. Engg. (CSE) January-June/2024
Course Code: CIT-311 Course Title: Microprocessor and Assembly Language
Credit Hour: 3.0 Full Marks: 15 Session: 2021-2022 Time: 01 Hour

1. Write down the steps to execute a machine instruction. Illustrate and describe the Intel 8086 Microprocessors architecture. 3
2. Define memory segment. Write down the features of 80286 microprocessor. 2
3. Write down the difference between physical and logical memory. A memory location has physical address 80FD2h. In what segment does it have offset BFD2h? 2
4. Determine the memory location addressed by the following real mode 80286 register combinations: DS = 1000H and DI = 2000H also draw the diagram of memory access. 2
5. How can you characterize a processor? Why minimum and maximum modes of operation used in 8086? 3
6. Distinguish between real and protected addressing mode. How does a segment register switch activities from real to protected? 3

Patuakhali Science and Technology University

B.Sc. Engg. (CSE) 5th Semester (Level-3, Semester-I) Mid Examination January-June 2024

Course code: CIT-315

Course Title: Artificial Intelligence

Session: 2019-20

Credit hour: 3.00

Full marks: 15

Duration: 1 hour

[Figures in the right margin indicate full marks. Split answering of any question is not recommended]

Answer the following questions.

1. (a) What is Artificial Intelligence (AI)? Briefly explain the difference between acting humanly, thinking humanly, acting rationally, and thinking rationally in AI. 02
- (b) A self-driving car's AI system detects a sudden, unavoidable obstacle and instantly swerves to avoid it, minimizing damage, but not eliminating it entirely. Is this action rational? Is it intelligent? How does this AI's "decision-making" differ from a human reflex, like pulling your hand from a hot stove 02
- (c) What is meant by the structure of an agent in AI? Explain how the interaction between agent architecture and agent program defines agent behavior. Distinguish between model-based reflex agents, and learning agents. 03
2. (a) What is an uninformed search? Distinguish the breadth first search, BFS, and the depth first search, DFS, algorithms. Compare the iterative deepening depth-first search, IDS, and BFS algorithms in terms of total number of nodes generated when the branching factor $b = 10$ and the depth $d = 5$. 04
- (b) Write the role of heuristic function in informed search strategy. How to minimize the total estimated solution cost using the best-first search, A* search, algorithm. Show the heuristic must be consistent for optimal solution in A* search algorithm. 04

Patuakhali Science and Technology University

5th Semester (Level-3, Semester-I), Mid Examination of B.Sc. Engg. (CSE), January-June: 2024

Course Code: CCE-313 Course Title: Computer Networks

Credit Hour: 3.0 Full Marks: 15

1. "Four levels of addresses are used in an internet employing the TCP/IP protocols"-justify the statement through a real time example. 3
2. How can you Extracting Information in a Block? 2
3. An address in a block is given as 190.100.50.25 as a classful address. Find the number of addresses in the block, the first address, usable address, broadcast address, and the last address. Also, draw the block of addresses. 3
4. What is the HOL blocking issue in HTTP/1.1? How does HTTP/2 attempt to solve it? 2
5.
 - i. For a DNS message, analyze the flag 0x8F80. Is it valid? Justify. 5
 - ii. A DNS client is looking for the IP address of xxx.yyy.com. Show the query message with values for each field.
 - iii. Show the response message of a DNS server to the query related to 5(ii). Assume the IP address is 201.34.23.12.

Logical
Port
Application

Full Marks: 15 Duration: 1 hour

Section A [Implement a,b or a,c]

1. a) Implement uniform-cost search and compare with breadth-first search. 08
b) Implement iterative deepening Search
c) Implement and prove that A* search is optimal if the heuristic function $f(n)$ is admissible and consistent.

Section B

1. You are given a propositional logic expression consisting of variables ($p, q, r, \dots$), logical operators (AND, OR, NOT, IMPLIES, IFF), and parentheses. 07

a Write a program that:

- Parses the logical expression.
- Evaluates its truth value for all possible truth assignments of the variables.
- Generates a truth table.

4. Extend your program to check whether the expression is:

- A **Tautology** (always true),
- A **Contradiction** (always false), or
- **Contingent** (true for some assignments and false for others).

Input Example: $(p \rightarrow q) \wedge (q \rightarrow r) \rightarrow (p \rightarrow r)$

2. Code the following proposition using PL with python and print the Truth Tables. 07

- a f. It is raining outside if and only if it is a cloudy day.
- b g. If you get a 100 on the final exam, then you earn an A in the class.
- c h. Take either 2 Advil or 3 Tylenol
- d i. She studied hard or she is extremely bright.
- e j. I am a rock and I am an island.

Course Code: CIT-312 Course Title: Microprocessor and Assembly Language Sessional

Part: 1 Credit Hour: 1.50 Full Marks:6 Duration: 45 Mins Id: 2102024

Write assembly language code, converted machine code, corresponding hex code, and mention the inputting procedure into KIT (like AD --> DA-->. . .) for the given assembly language code segment.

AX ← E

$\left\langle \frac{3x^2}{2y} - 2; \text{ where the value of } x = E \quad y = A \right\rangle$

95/ 4 5

Course Code: CIT-312 Course Title: Microprocessor and Assembly Language Sessional

Part: 1 Credit Hour: 1.50 Full Marks:15 Duration: 45 Mins

Set-E Write assembly language code, converted machine code, corresponding hex code, and mention the inputting procedure into KIT (like AD --> DA-->. . .) for the given assembly language code segment

$$\frac{3X}{2Y} - \frac{X}{Y}; \text{ where the value of } x = E \quad y = A$$

Course Code: CIT-312 Course Title: Microprocessor and Assembly Language Sessional

Part: 1 Credit Hour: 1.50 Full Marks:15 Duration: 45 Mins

Set-F Write assembly language code, converted machine code, corresponding hex code, and mention the inputting procedure into KIT (like AD --> DA-->. . .) for the given assembly language code segment

$$\frac{5X}{2Y} + \frac{X}{Y}; \text{ where the value of } x = E \quad y = 8$$

Mar 2+2+1+1=6

Course Code: CTT-312 Course Title: Microprocessor and Assembly Language Sessional

Part: 1 Credit Hour: 1.50 Full Marks:15 Duration: 45 Mins

Set-C Write assembly language code, converted machine code, corresponding hex code, and mention the inputting procedure into KIT (like AD --> DA--> . . .) for the given assembly language code segment

$$\frac{3x^2}{2x+1}; \text{ where the value of } x = E \quad y = A$$